GUILHERME HENRIQUE RISSON SILVA

AI Software Engineer | Backend & Data | Python | PostgreSQL | FastAPI | LLM
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SUMMARY

Mid-level data analyst at Vitru Educação, where I build the Python systems behind UniCesumar's academic planning. The main one is the faculty allocation optimizer (OR-Tools, FastAPI, React), which assigns **about 200 instructors to more than 350 classes per module in minutes**. In open-source projects I build LLM applications: RAG with hybrid search, document data extraction with FastAPI, and an agent that publishes videos with checked sources.

EXPERIENCE

Vitru Educação (UniCesumar) Hybrid, Maringá, Brazil
Mid-level Data Analyst, Academic Planning 03/2026 – Present

- Built the **faculty allocation optimizer** (Python, OR-Tools CP-SAT, FastAPI, React, TypeScript), which assigns about 200 instructors to more than 350 classes per module **in minutes**, work that used to take weeks in spreadsheets.
- Built the 2027 **student enrollment forecasting model** (Python, DuckDB, Parquet, Power BI), with 50 automated quality checks and **error below 2%** against actuals.
- Maintain in production the **weekly curriculum pipeline** for UniCesumar and Uniasselvi (Python, SQL Server), which reads 6 spreadsheet layouts through 4 validation layers.
- Moved the **planned-vs-actual audits** from Power Query to Python, over datasets of up to **40 million rows**, with Power BI dashboards for managers and leadership.
- Rewrote in Python and Marimo the **lab class demand vs. tutor capacity model**, with a simulator by campus and city whose scenarios cut the shortfall from **27,400 to 21,300 hours**.

Linoforte Móveis Osvaldo Cruz, Brazil
Assistant Operator 06/2022 – 12/2025

- Set up monitoring of **production and quality KPIs**, flagging recurring stoppages for corrective action.
- Mapped bottlenecks and downtime and proposed improvements that **reduced rework and waste** on the line.

PROJECTS (github.com/guilhermerissondev)

- **Regulatory RAG Eval:** question answering over 294 Brazilian Central Bank regulations with source citations (Python, BM25, embeddings, Gemini). Hybrid search raised **MRR from 0.361 to 0.644** over keyword search.
- **Procurement Triage Service:** API that uses an LLM to extract data from Brazilian public procurement notices (PNCP) and only accepts values found in the PDF (Python, FastAPI, Gemini, Docker). **US$ 0.028 per notice**, 61 tests in CI.
- **TikTok Viral Generator:** agent that tracks 8 news sources, researches, scripts and publishes short videos with a source for every fact (Python, Pydantic, SQLite). Publishes **4 videos a day**, with 691 tests.
- **Silent Data Defects:** command-line tool that profiles data quality in large datasets (Python, DuckDB, PyArrow). Processes 5 million rows in 4.8 s, **25x faster** than ydata-profiling.
- **Credit Risk Decision Engine:** credit risk model with the approval cutoff set by expected profit (Python, LightGBM, SHAP). **+24% profit** with the same model.

EDUCATION

B.S. in Software Engineering — Uninter 10/2026 – 2030 (in progress)
AI Software Engineer 4.0 Program (360 h) — Data Science Academy 09/2026 – 12/2026 (expected)
Postgraduate Certificate in Data Science (360 h) — UniCesumar 03/2026 – 10/2026
B.S. in Civil Engineering — FAI, Faculdades Adamantinenses Integradas 2021 – 2025

SKILLS

Languages: Python, SQL, TypeScript
AI and LLM: RAG, hybrid search (BM25 + embeddings), structured output, LLM evaluation, agents, Gemini API
Backend: FastAPI, Pydantic, React, Docker, pytest, GitHub Actions, Git
Data and ML: pandas, DuckDB, Parquet, SQL Server, scikit-learn, LightGBM, OR-Tools (CP-SAT), Power BI
Spoken languages: Portuguese (native), English (professional), Spanish (intermediate)